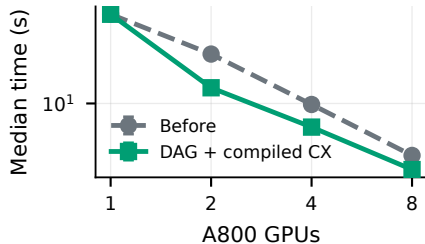
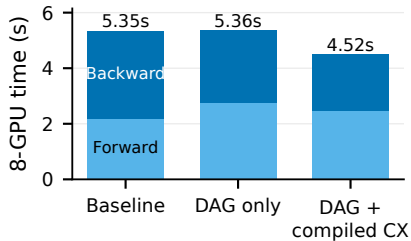


Ring needs communication-aware scheduling and compiled execution together

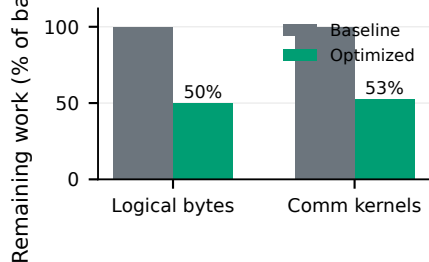
a Ring strong scaling



b DAG alone: no speedup



c Half the communication



31 qubits · 8 layers · 248 trainable parameters · full value + gradient · warmup 2 + measured 5 · 16-GPU rerun pending